Pages
Date 03 02 25

THEORY OF COMPUTATION

UNIT-1

Introduction of Automata Theory

MODELS:

\# Finite Automata → language acceptor

- → F.A. with o/p: after accepting generate o/p
  - → Mealy m/c
  - → Moore m/c
- → F.A. without o/p: only accept / reject
  - → DFA (Deterministic FA)
  - → NFA (Non-deterministic FA)

$\rightarrow (q_0) \xrightarrow{0} ((q_1))$

from 1 state with one i/p can only move once

$(q_0) \xrightarrow{0} q_0$

$(q_0) \xrightarrow{1} q_1$

from 1 state with one i/p can have more than 1 option to move.

- Symbol: is a basic building block which can be any character / token.

- Alphabet: is a finite non-empty set of symbols (every lang. has its own alphabet)
  eg: $\Sigma = \{0, 1\}$

• String: is a finite sequence of symbols (which we have no. of set alphabet

$\Sigma = \{0, 1\}$

$= \{0, 1, 00, 11, 01, 10, \ldots\}$

↳ String (subset)

• Formal Language: is a define as a set of strings. we apply these model from words to sentences.

$L = \{a^n b^n \mid n \geq 1\} = \{ab, aabb, aaabbb, \ldots\}$

$\Sigma = \{a, b\}$ ∴

alphabet

Introduction:

An automata is an abstract model of digital computer, an automation has a mechanism to read input from the i/p tape.

• Any language is recognised by some automata.

• These are basically lang. acceptor or lang. recogniser.

• Regular langs. are accepted by the finite automata

Formal defn of finite automata

It is the combination of 5 tuples:

$(Q, \Sigma, \delta, q_0, F)$

Q = finite set of states

Σ = finite set of i/p symbol

q_0 = initial state → always 1

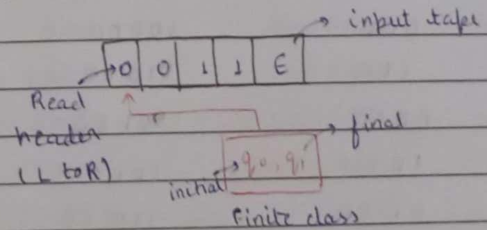
F = set of final states → can be more than 1

δ = transition function or mapping function

what will be the next state based on q will be determined by transition function $\left\{ \delta(q, w) \vdash^* (q) \right.$

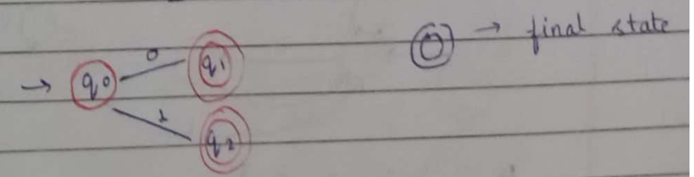
state ↳ String ↳ Next state

Finite Automata Model



DFA (Deterministic Finite Automata)

$\Sigma = (0, 1)$



The deterministic finite automata is deterministic in nature, each move in these automata is uniquely define on current i/p & current state.

Def^n: It is a combination of 5 tuples
$(Q, \Sigma, \delta, q_0, F)$
where
every things are same except δ

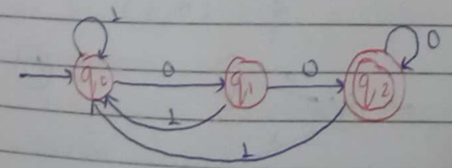
$\delta : (Q \times \Sigma) \rightarrow Q$

$Q = 3\ (q_0, q_1, q_2)$

Q1: Construct a DFA for a language which ends from 00 where $\Sigma = \{0,1\}$

$\Sigma = \{0,1\}$

00	00000
000	10000
100	01000
0000	00100
1000	10100
0100	11000
1100	01100
	11100



$Q = \{q_0, q_1, q_2\}$
$\Sigma = \{0,1\}$
$q_0 = q_0$
$F = \{q_2\}$

$\delta = Q \backslash \Sigma$	0	1
$\rightarrow q_0$	$\{q_1\}$	$\{q_0\}$
q_1	$\{q_2\}$	$\{q_0\}$
q_2	$\{q_2\}$	$\{q_0\}$

$\delta(q_0, 00) \vdash (q_1, 0)$
$\vdash (q_2)$

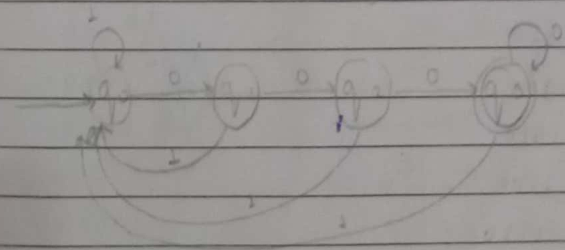
Hence, q_2 is final state so, that is accepted by this m/c.

06/02/25

Q2: Construct a DFA for a language ending with 000 where ~~[illegible]~~ $\Sigma = \{0,1\}$

$\Sigma = \{0,1\}$

000	000000	0001000
1000	010000	
0000	100000	
11000	001000	
01000	110000	
10000	101000	
00000	011000	
	111000	



$Q = \{q_0, q_1, q_2, q_3\}$
$\Sigma = \{0, 1\}$
$q_0 = q_0$
$F = \{q_3\}$

$\delta = Q \backslash \Sigma$	0	1
q_0	q_1	q_0
q_1	q_2	q_0
q_2	q_3	q_0
q_3	q_3	q_0

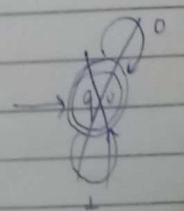
String pass example:

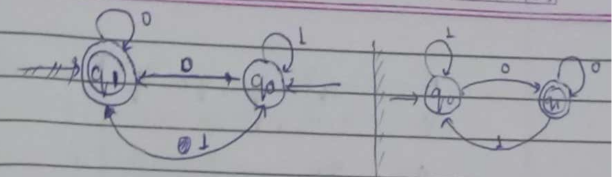
$\delta(q_0, 01000) \vdash (q_1, 1000)$
$\vdash (q_0, 000)$
$\vdash (q_1, 00)$
$\vdash (q_2, 0)$
$\vdash (q_3)$

Q3. Construct a DFA for a binary no. which is divisible by 2.

$\Sigma = \{0, 1\}$

0 (for divisible by 2) → in last 0 place must come
10 11000
100 11010
110
1000
1010
1100
1110
10000
10010





$Q = \{q_0, q_1\}$
$\Sigma = \{0, 1\}$
$q_0 = q_0$
$F = \{q_1\}$

Here, we cannot make q_0 as final state bc a/c to Que we must pass i/p to check it's divisibility with 2.

$\delta = Q \backslash \Sigma$	0	1
q_0	q_1	q_0
q_1	q_1	q_0

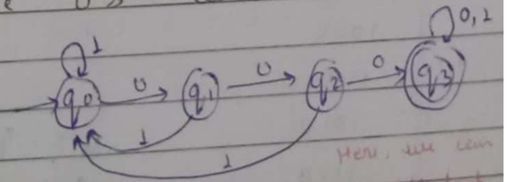
String pass example:

$\delta(q_0, 11010) \vdash (q_0, 1010)$
$\vdash (q_0, 010)$
$\vdash (q_1, 10)$
$\vdash (q_0, 0)$
$\vdash (q_1)$

Hence, q_1 is final state so, that string is accepted by this m/c.

Q3 Construct a DFA for a language has 3 consecutive 0's where $\Sigma = \{0, 1\}$

000
1000
0001
11000
00011
10001
1101000
10111000
1010101000



Here, we can do self loop of 1 mcz a/c to Que it is not specified that it must end with 0 only

$\delta = Q \backslash \Sigma$	0	1
q_0	q_1	q_0
q_1	q_2	q_0
q_2	q_3	q_0
q_3	q_3	q_3

$Q = \{q_0, q_1, q_2, q_3\}$
$\Sigma = \{0, 1\}$
$q_0 \; q_0$
$qF = \{q_3\}$

String pass example

$\delta(q_0, 010001) \vdash (q_1, 10001)$
$\vdash (q_0, 0001)$
$\vdash (q_1, 001)$
$\vdash (q_2, 01)$
$\vdash (q_3, 1)$
$\vdash (q_3)$

Hence, q_3 is final state ~~so~~, that string is accepted by this m/c

Q 4. Construct a DFA which accept only those strings which start with 1 & ends with 0: $\Sigma = \{0,1\}$

10
110
100
1000
1100
1010
1110

$Q = \{q_0, q_1, q_2\}$
$\Sigma = \{0, 1\}$
$q_0 = q_0$
$qF = \{q_2\}$

δ:

$Q \backslash \Sigma$	0	1
q_0	-	q_1
q_1	q_2	q_1
q_2	q_2	q_1

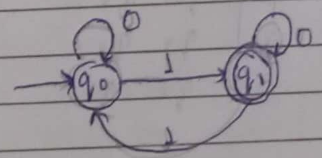
String pass example

$\delta(q_0, 1100) \vdash (q_1, 100)$
$\vdash (q_1, 00)$
$\vdash (q_2, 0)$
$\vdash (q_2)$

Hence, q_2 is the final state so, that string is accepted by this m/c

Q5. Construct a DFA which accepts odd number of 1's and any no. of 0's.

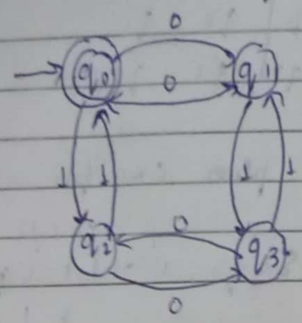
$\Sigma = \{0,1\}$



1
10
01
~~111~~
111
100
001
11111
10101
010101
101010

Design a DFA which accepts even no. of 0's & even no. of 1's.

11
00
0011
1100
1010
0101
0110
101101
010010



Here, q_0 can final state bcz we can pass empty i/p initially also

$Q = \{q_0, q_1, q_2, q_3\}$
$\Sigma = \{0, 1\}$
$q_0 = q_0$
$F = \{q_0\}$

$\delta =$

$Q \backslash \Sigma$	0	1
q_0	q_1	q_2
q_1	q_0	q_3
q_2	q_3	q_0
q_3	q_2	q_1

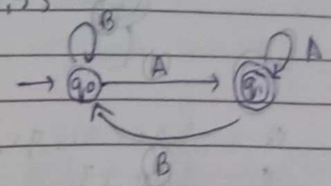
String pass example
$\delta(q_0, 101101) \vdash (q_2, 01101)$
$\vdash (q_3, 1101)$
$\vdash (q_1, 101)$
$\vdash (q_3, 01)$
$\vdash (q_2, 1)$
$\vdash (q_0)$

Hence, q_0 is the final state so, that string is accepted by m/c

Q Construct a DFA for divisible by 2 for decimal number

$\Sigma = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

$A = \{0, 2, 4, 6, 8\}$
$B = \{1, 3, 5, 7, 9\}$



$Q = \{q_0, q_1\}$
$\Sigma = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$
$q_0 = q_0$
$F = \{q_1\}$

$\delta =$

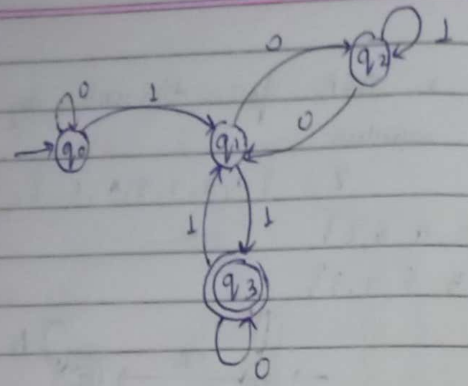
$Q \backslash \Sigma$	A	B
q_0	q_1	q_0
q_1	q_1	q_0

String pass example.
$\delta(q_0, 24) \vdash (q_0, 4)$
$\vdash (q_1)$

Hence, q_1 is the final state so, that string is accepted by m/c

Q Construct a DFA for binary number which is divisible by 3.

11 10101 21
110 11000 24
1001 9
1100 12
10010 18



$Q = \{q_0, q_1, q_2, q_3\}$

$\Sigma = \{0, 1\}$

$q_0 = q_0$

$F = \{q_3\}$

δ =

$Q \backslash \Sigma$	0	1
q_0	q_0	q_1
q_1	q_2	q_3
q_2	q_1	q_2
q_3	q_3	q_1

String pass example

$\delta(q_0, 11011) \vdash (q_1, 1011)$

$\vdash (q_3, 011)$

$\vdash (q_3, 11)$

$\vdash (q_1, 1)$

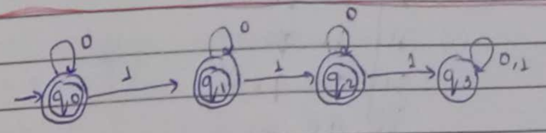
$\vdash (q_3)$.

Hence, q_3 is the final state so that string is accepted by the m/c.

Q. Design a DFA for a machine M where $L = \{\omega / \omega \in (0,1)^*\}$ & ω is a string that does not contain 3 consecutive 1's.

$(0,1)^* \rightarrow$ Kleen closure

↳ $\{\epsilon, 0, 1, 00, 11, 01, 10, 000, 001, 010, 011, 100, 101, 110, ~~111~~, \ldots\}$



(for a string that does not contain 3 1's)

$Q = \{q_0, q_1, q_2, q_3\}$

$\Sigma = \{0, 1\}$

$q_0 = q_0$

$F = \{q_0, q_1, q_2\}$

δ =

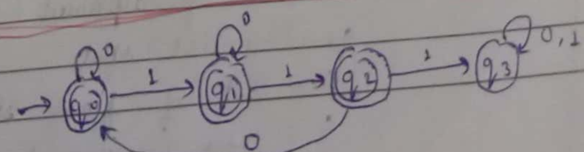
$Q \backslash \Sigma$	0	1
q_0	q_0	q_1
q_1	q_1	q_2
q_2	q_2	q_3
q_3	q_3	q_3

String pass example

$\delta(q_0, 110) \vdash (q_1, 10)$

$\vdash (q_2, 0)$

$\vdash (q_2)$



Q : $\{q_0, q_1, q_2, q_3\}$
Σ : $\{0, 1\}$
q_0 : q_0
F : $\{q_0, q_1, q_2\}$

δ = $Q \backslash \Sigma$	0	1
q_0	q_0	q_1
q_1	q_1	q_2
q_2	q_0	q_3
q_3	q_3	q_3

$\delta(q_0, 11011) \vdash (q_1, 1011)$
$\vdash (q_2, 011)$
$\vdash (q_0, 11)$
$\vdash (q_1, 1)$
$\vdash (q_2)$

Non Deterministic Finite Automata:
The finite automata is called NFA when there exist more than one path for a specific input from current state to next state.

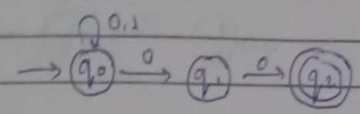
It is a collection of 5 tuples
$M = \{Q, \Sigma, q_0, F, \delta\}$
Q = Set of finite string
Σ = Set of alphabet / i/p symbol
q_0 = initial state
F = set of final state
δ = ~~Map~~ Transition funct^n / Mapping

NFA = $Q \times \Sigma \rightarrow 2^Q$

13/2/25

Q Construct a NFA for a language ending with 00 where $\Sigma = \{0, 1\}$

00
000
100
0000
1000
0100
1100



after separating the minimum condition (00) we get
$L = \{0, 1\}^*$ $\{\epsilon, 0, 1, 00, 01, 01, 0,$

$q_0 \xrightarrow{0} q_0 \xrightarrow{0} q_0$ for 00
$q_0 \xrightarrow{0} q_1 \xrightarrow{0} q_2$
we have 2 path for every input 1 or 0.

Regular expression : $(0, 1)^* 00$

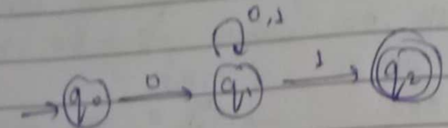
Tuples :- $Q = \{q_0, q_1, q_2\}$
$\Sigma = \{0, 1\}$
$q_0 = q_0$
$F = \{q_2\}$

$Q \times \Sigma = 2^Q$
$3 \times 2 = 2^3$

δ = $Q \backslash \Sigma$	0	1
ϕ	ϕ	ϕ
q_0	$\{q_0, q_1\}$	q_0
q_1	q_2	—
q_2	—	—
$q_0 q_1$	$\{q_0 q_1 q_2\}$	$\{q_0\}$
$q_0 q_2$	$\{q_0 q_1\}$	$\{q_0\}$
$q_1 q_2$	$\{q_2\}$	—
$q_0 q_1 q_2$	$\{q_0 q_1 q_2\}$	$\{q_0\}$

Q. Construct a NFA for a lang. start with 0 & end with 1 & $\Sigma = \{0,1\}$

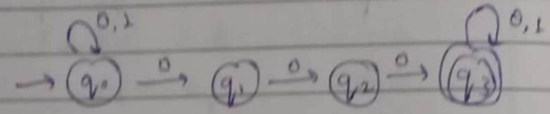
01
001
011
0001
0011
0101
0111
00001
00011
00101
00111
01001



graphical representation
$= 0(0,1)^* 1$

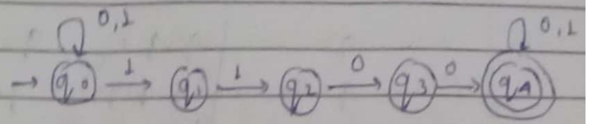
Q. Construct a NFA where $\Sigma = \{0,1\}$ for 3 consecutive 0's

000
0000
1000
00000
01000
10000
11000
10001
00011



Q. Design NFA which accepts the string 11 is followed by 00 where $\Sigma = \{0,1\}$

1100
01100
11100
001100
011100
101100

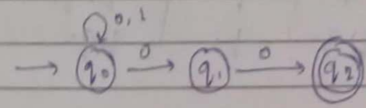


Conversion of NFA to DFA

Q. Construct a NFA for L: ending with 00, where $\Sigma = (0,1)$

NFA: $M : (Q, \Sigma, q_0, F, \delta)$
$\hookrightarrow Q \times \Sigma \to 2^Q$

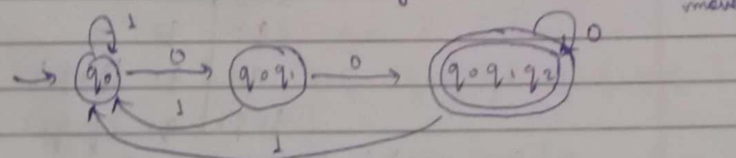
main m/c



NFA to DFA

	δ	0	1
states for DFA	q_0	$(q_0 q_1)$	q_0
	$q_0 q_1$	$q_0 q_1 q_2$	q_0
	$q_0 q_1 q_2$	$q_0 q_1 q_2$	q_0

(take every new state & map its move)

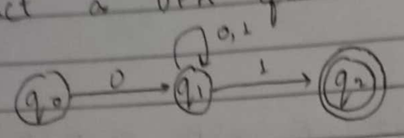


Tuples for M' = $Q' = \{q_0, q_0q_1, q_0q_1q_2\}$
$\Sigma' = \{0,1\}$
$q_0' = q_0$
$F' = q_0q_1q_2$

	0	1
q_0	(q_0q_1)	q_0
q_0q_1	$(q_0q_1q_2)$	q_0
$q_0q_1q_2$	$(q_0q_1q_2)$	q_0

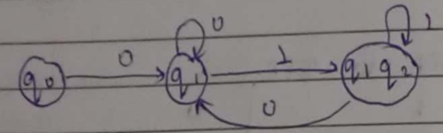
Q Construct a DFA from NFA

M :



NFA to DFA

	0	1
initial state → q_0	q_1	–
q_1	q_1	q_1q_2
final state → q_1q_2	q_1	q_1q_2



Tuples for M' : $Q' = \{q_0, q_1, q_1q_2\}$

$\Sigma' = \{0, 1\}$

$q_0' = q_0$

$F' = q_1q_2$

δ' =

	0	1
q_0	q_1	–
q_1	q_1	(q_1q_2)
q_1q_2	q_1	(q_1q_2)

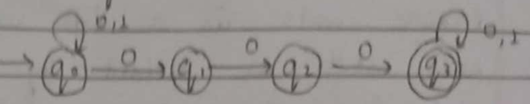
$\delta(q_0, 011) \vdash (q_1, 11)$

$\vdash ((q_1q_2), 1)$

$\vdash (q_1q_2)$

Q NFA to DFA for 3 consecutive 0's

M :→



write tuples for both M & M'

For M, Tuples:

$M = \{Q, \Sigma, q_0, F, \delta\}$

$Q = \{q_0, q_1, q_2, q_3\}$

$\Sigma = \{0, 1\}$

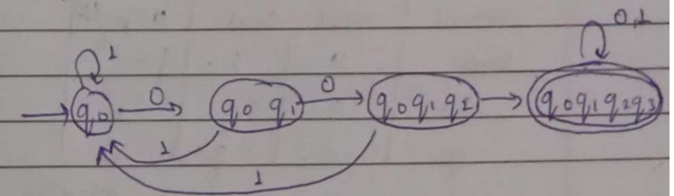
$q_0 = q_0$

$F = \{q_3\}$

δ :

	0	1
q_0	(q_0q_1)	q_0
q_1	q_2	–
q_2	q_3	–
q_3	q_3	q_3

M' :→



For M' Tuples:

$M' = \{Q', \Sigma', q_0', F', \delta'\}$

$Q' = \{q_0, q_0q_1, q_0q_1q_2, q_0q_1q_2q_3\}$

$\Sigma' = \{0, 1\}$

$q_0' = q_0$

$F = \{q_0q_1q_2q_3\}$

δ :

	0	1
q_0	(q_0q_1)	q_0
q_0q_1	$(q_0q_1q_2)$	q_0
$q_0q_1q_2$	$(q_0q_1q_2q_3)$	q_0
$q_0q_1q_2q_3$	$(q_0q_1q_2q_3)$	$(q_0q_1q_2q_3)$

Moore Machine:

Moore m/c is a FA in which the next state is decided by the current state & current input symbol.

The output symbol at a given time depends only on the present state of m/c.

Moore m/c is a combination of 6 tuples

$(Q, \Sigma, q_0, \delta, \Delta, \lambda)$

Q : set of finite states

Σ = set of input symbol

q_0 : initial state

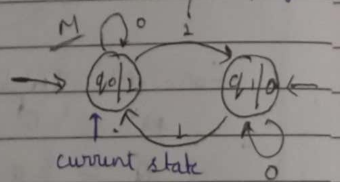
δ = mapping funct^n / transition funct^n

Δ = it is an output alphabet

λ = it is an output funct^n $(Q \to \Delta)$

δ :

Current State	Next State 0	Next State 1	O/p (λ) funct^n
q_0	q_0	q_1	1
q_1	q_1	q_0	0



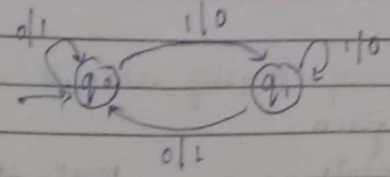
Length i/p string : 0100 = 4

(n+1) o/p string = 11000 = 5

17/2/25

Mealy Machine:

In mealy m/c output symbol depend upon present input symbol & present state of the m/c.



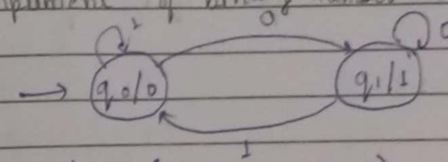
$M : (Q, \Sigma, q_0, \delta, \Delta, \lambda)$

Moore : $\lambda \to Q \to \Delta$

Mealy $\lambda \to Q \times \Sigma \to \Delta$

Q. Design a Moore m/c to generate 1's complement of binary number

$\Sigma = \{0, 1\}$



$M = (Q, \Sigma, q_0, \delta, \Delta, \lambda)$

$Q = \{q_0, q_1\}$

$\Sigma = \{0, 1\}$

$q_0 : q_0$

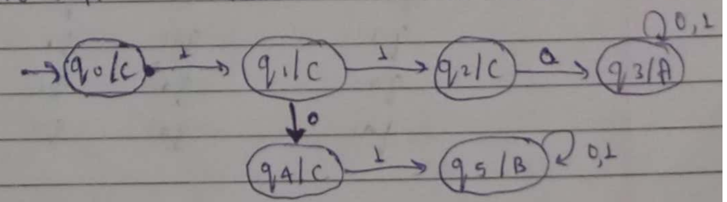
δ :

CS	NS 0	NS 1	λ
q_0	q_1	q_0	0
q_1	q_1	q_0	1

Q. Design a Moore m/c for binary input such that if input substring is A → 101 then output is B otherwise C

19/2/25

110 → A 101 → B otherwise → C



Q Conversion of Mealy m/c to moore m/c

Step 1 : given

Current State	Next State 0	Δ	1	Δ
q_1	q_3	0	q_2	0
q_2	q_1	1	q_4	0
q_3	q_2	1	q_1	1
q_4	q_4	1	q_3	0

$q_1 = 1$

$q_2 = (0,1)$ — $q_2 \rightarrow q_{20}, q_{21}$

$q_3 = 0$

$q_4 = (0,1)$ — $q_4 \rightarrow q_{40}, q_{41}$

Step 2 : Current State / Next State

Current State	0	Δ	1	Δ
q_1	q_3	0	q_{20}	0
q_{20}	q_1	1	q_{40}	0
q_{21}	q_1	1	q_{40}	0
q_3	q_{21}	1	q_1	1
q_{40}	q_{41}	1	q_3	0
q_{41}	q_{41}	1	q_3	0

Step 3 : Current State / Next State / λ

Current State	0	1	λ
q_1	q_3	q_{20}	1
q_{20}	q_1	q_{40}	0
q_{21}	q_1	q_{40}	1
q_3	q_{21}	q_1	0
q_{40}	q_{41}	q_3	0
q_{41}	q_{41}	q_3	1

$Q = \{q_1, q_{20}, q_{21}, q_3, q_{40}, q_{41}\}$

$\Sigma = \{0, 1\}$

$q_0 = q_1$

Q Current State / Next State

Current State	0	Δ	1	Δ
q_1	q_1	1	q_2	0
q_2	q_4	1	q_4	1
q_3	q_2	1	q_3	1
q_4	q_3	0	q_1	1

$q_1 = 1$

$q_2 = \{0,1\}$ — q_{20} q_{21}

$q_3 = \{0,1\}$ — q_{30} q_{31}

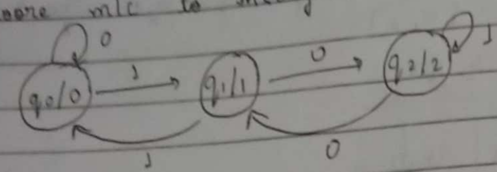
$q_4 = 1$

Current State	0	Δ	1	Δ
q_1	q_1	1	q_{20}	0
q_{20}	q_4	1	q_4	1
q_{21}	q_4	1	q_4	1
q_{30}	q_{21}	1	q_{31}	1
q_{31}	q_{21}	1	q_{31}	1
q_4	q_{30}	0	q_1	1

Current State	0	1	λ
q_1	q_1	q_{20}	1
q_{20}	q_4	q_4	0
q_{21}	q_4	q_4	1
q_{30}	q_{21}	q_{31}	0
q_{31}	q_{21}	q_{31}	1
q_4	q_{30}	q_1	1

Date 10 2 25

Moore m/c to Mealy m/c.

Q 

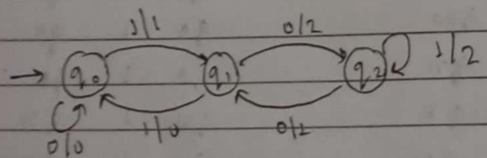
Step 1:

δ :

C.S	NS 0	NS 1	λ
q_0	q_0	q_1	0
q_1	q_2	q_0	1
q_2	q_1	q_2	2

Step 2:

CS	Next State 0	Δ	1	Δ
q_0	q_0	0	q_1	1
q_1	q_2	2	q_0	0
q_2	q_1	1	q_2	2

Q 

← Mealy m/c to moore m/c

Transition Table:

CS	Next State 0	Δ	1	Δ
q_0	q_0	0	q_1	1
q_1	q_2	2	q_0	0
q_2	q_1	1	q_2	2

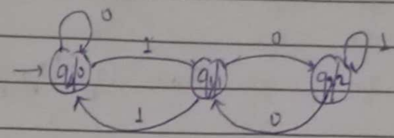
q_0 : 0

q_1 : 1

q_2 : 2

Pages
Date

C.S	0	1	λ
q_0	q_0	q_1	0
q_1	q_2	q_0	1
q_2	q_1	q_2	2



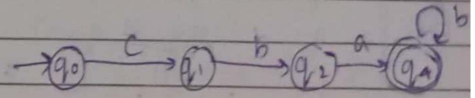
Q Construct a NFA for $L = (cbab^n \mid n \geq 0)$

cba

cbab

cbabb

cbabbb



Q Design DFA that accepts all string with at most 3 a , $\Sigma = \{a, b\}$

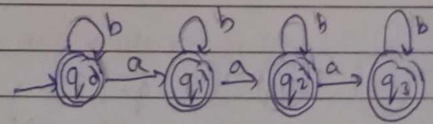
b

ab

aba

aaa

aaaa X



Date: 9.1.2.25

UNIT - 2

Minimization of DFA

Q/Σ	0	1
I.S. (initial state) → q_0	q_1	q_5
q_1	q_6	q_2
F.S. (final state) → q_2	q_0	q_2
q_3	q_2	q_6
q_4	q_7	q_5
q_5	q_2	q_6
q_6	q_6	q_4
q_7	q_6	q_2

$Q_1^0 = (F)$

$Q_2^0 = (Q - Q_1)$

set 1 set 2

$\pi_0 = \{(q_2)(q_0 q_1 q_3 q_4 q_5 q_6 q_7)\}$

$= q_0\{q_0, q_4, q_6\} \neq q_0\{q_1 q_3 q_5 ...\}$

$\pi_1 = \{(q_2)(q_0 q_4 q_6)(q_1 q_3 q_5 q_7)\}$

$= q_0\{q_0 q_4\} \neq q_0\{q_6\}$

$= q_1\{q_1, q_7\} \neq q_1\{q_3 q_5\}$

{both in same sets} ✓

$q_0 \xrightarrow{0} (q_1)$, $q_1 \xrightarrow{0} (q_6)$ ×

$q_0 \xrightarrow{1} (q_5)$, $q_1 \xrightarrow{1} (q_2)$

× $q_0 \xrightarrow{0} (q_1)$, $q_4 \xrightarrow{0} (q_2)$ — q_0, q_4 — q_2 not in same set

$\pi_2 = \{(q_2)(q_0 q_4)(q_1 q_7)(q_3 q_5)(...\}$

$\pi_3 = \{(q_2)(q_0 q_4)(q_1 q_7)(q_3 q_5)(q_6)\}$

$\pi_0, \pi_1 \ldots \pi_n$

check till $\pi_{n-1} = \pi_n$

$M' = \{Q, \Sigma, \delta, q_0, F\}$

$Q = \{(q_2)(q_0 q_4)(q_1 q_7)(q_3 q_5)(q_6)\}$

$\Sigma = \{0, 1\}$

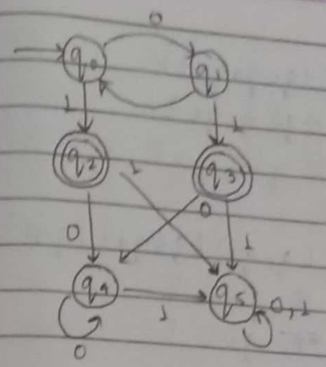
$q_0 = q_0$

$F = \{q_2\}$

δ =

Q\Σ	0	1
→ $q_0 q_4$	$(q_1 q_7)$	$(q_3 q_5)$
$q_1 q_7$	q_6	q_2
q_2	$(q_0 q_4)$	q_2
$q_3 q_5$	(q_2)	q_6
q_6	q_6	$(q_0 q_4)$

Q



	0	1
→ q_0	q_1	q_2
q_1	q_0	q_3
q_2	q_4	q_5
q_3	q_4	q_5
q_4	q_4	q_5
q_5	q_5	q_5

$Q_1^0 = F = \{q_2 q_3\}$

$Q_2^0 = (Q - Q_1) = \{q_0 q_1 q_4 q_5\}$

$\pi_0 = \{(q_2 q_3)(q_0 q_1 q_4 q_5)\}$

$= q_2\{q_2\ q_3\} \neq q_2\{\ \}$

$= q_0\{q_0\ q_1\} \neq q_0\{q_4\ q_5\}$

$\pi_1 = \{(q_2 q_3)(q_0 q_1)(q_4 q_5)\}$

$= q_0\{q_0\ q_1\} \neq q_0\{\ \}$

$= q_4\{q_4\ q_5\} \neq q_4\{\ \}$

$\pi_2 = \{(q_2 q_3)(q_0 q_1)(q_4 q_5)\}$

Q

Q\Σ	0	1
→ q_0	q_1	q_0
q_1	q_0	q_2
q_2	q_3	q_1
q_3	q_3	q_0
q_4	q_3	q_5
q_5	q_6	q_4
q_6	q_5	q_6
q_7	q_6	q_3

$Q_1^0 = F = \{q_3\}$

$Q_2^0 = Q - Q_1$

$= \{q_0 q_1 q_2 q_4 q_5 q_6 q_7\}$

$\pi_0 = \{(q_3)(q_0 q_1 q_2 q_4 q_5 q_6 q_7)\}$

$= q_0\{q_0 q_1 q_5 q_6\} \neq q_0\{q_2 q_4 q_7\}$

$\pi_1 = \{(q_3)(q_0 q_1 q_5 q_6)(q_2 q_4 q_7)\}$

$= q_0 \{q_0 q_6\} \neq q_0 \{q_1 q_5\}$

$= q_2 \{q_2 q_4\} \neq q_2 \{q_7\}$

$\pi_2 = \{(q_3)(q_0 q_6)(q_1 q_5)(q_2 q_4)(q_7)\}$

$= q_0 \{q_0 q_6\} \neq q_0 \{\}$

$= q_1 \{q_1 q_5\} \neq q_1 \{\}$

$= q_2 \{q_2 q_4\} \neq q_2 \{\}$

$\pi_3 = \{(q_3)(q_0 q_6)(q_1 q_5)(q_2 q_4)(q_7)\}$

$M' = \{Q, \Sigma, q_0, F, \delta\}$

$Q = \{(q_3)(q_0 q_6)(q_1 q_5)(q_2 q_4)(q_7)\}$

$\Sigma = \{0, 1\}$

$q_0 = q_0$

$F = q_3$

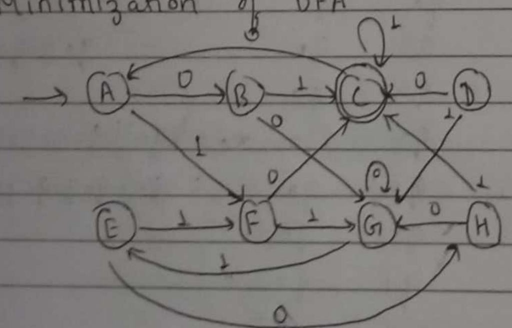
$\delta =$

$Q \backslash \Sigma$	0	1
$q_0 q_6$	$q_1 q_5$	$q_0 q_6$
$q_1 q_5$	$q_0 q_6$	$q_2 q_4$
$q_2 q_4$	q_3	$q_1 q_5$
q_3	q_3	$q_0 q_6$
q_7	$q_0 q_6$	q_3

14/2/25

Minimization of DFA

Q.



	0	1
→A	B	F
B	G	C
C	A	C
D	C	G
E	H	F
F	C	G
G	G	E
H	G	C

$Q_0' = F = \{C\}$

$Q_1' = Q - Q = \{A, B, D, E, F, G, H\}$

$\pi_0 = \{(C)(ABDEFGH)\}$

$= A\{AEG\} \neq A\{BDFH\}$

$\pi_1 = \{(C)(AEG)(BDFH)\}$

$= A\{AE\} \neq A\{G\}$

$= B\{BH\} \neq B\{DF\}$

$\pi_2 = \{(C)(AE)(BH)(DF)(G)\}$

$= A\{AE\} \neq A\{\}$

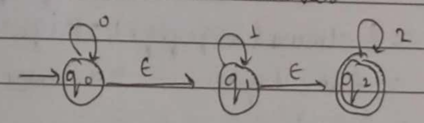
$= B\{BH\} \neq B\{\}$

$= D\{DF\} \neq D\{\}$

$\pi_3 = \{(AE)(BH)(C)(DF)(G)\}$

$\therefore \pi_3 = \pi_2$

NFA with ∈ moves



Regular expression $= 0^* 1^* 2^*$

$\delta = Q \times \Sigma \to 2^Q$

$= Q \times (\Sigma \cup \epsilon) \to 2^Q$

ϵ closure of $q_0 = \{q_0, q_1, q_2\}$

ϵ closure of $q_1 = \{q_1, q_2\}$

ϵ closure of $q_2 = \{q_2\}$

String
ε
0
1
00
10
01
11

Ye samak

δ :

	0	1	2	ε
→ q_0	q_0	–	–	q_1
q_1	–	q_1	–	q_2
q_2	–	–	q_2	–

0 1 2
→ q_0
q_1
q_2

$\delta(q_0, 0) = \epsilon$ closure $(\delta(\epsilon$ closure of $(q_0), 0)$
$= \epsilon$ closure $(\delta(q_0 q_1 q_2), 0)$
$= (q_0, 0) \cup (q_1, 0) \cup (q_2, 0)$
$= q_0 \cup - \cup -$
$= \epsilon$ closure of (q_0)
$= (q_0, q_1, q_2)$

$\delta(q_0, 1) = ((q_0 q_1 q_2), 1) = (q_0, 1) \cup (q_1, 1) \cup (q_2, 1)$
$= - \cup q_1 \cup - = \epsilon$ closure of $\{q_1\}$
$= \{q_1, q_2\}$

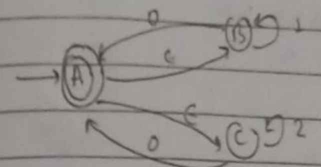
$\delta(q_0, 2) = \epsilon$ closure $(\delta(\epsilon$ closure of $(q_0), 2))$
$= \epsilon$ closure $(\delta(q_0 q_1 q_2), 2) = (q_0, 2) \cup (q_1, 2) \cup (q_2, 2)$
$= - \cup - \cup q_2$ $= \epsilon$ closure of $\{q_2\}$
$= \{q_2\}$

$\delta(q_1, 0) = ((q_1 q_2), 0) = (q_1, 0) \cup (q_2, 0) = - \cup -$
$= \epsilon$ closure of $- = -$

$\delta(q_1, 1) = \epsilon$ closure of $(\delta(\epsilon$ closure of $(q_1), 1))$
$= \epsilon$ closure of $(q_1 q_2, 1) = (q_1, 1) \cup (q_2, 1)$
$= q_1 \cup -$ $= \epsilon$ closure of $(q_1) = \{q_1, q_2\}$

	0	1	2
q_0	$q_0 q_1 q_2$	$q_1 q_2$	q_2
q_1	–	$q_1 q_2$	q_2
q_2	–	–	q_2

Q



ε closure of A = {A, B, C}
ε closure of B = {B}
ε closure of C = {C}

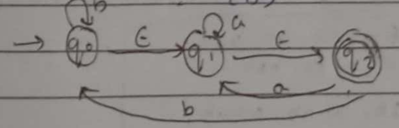
	0	1	2	ε
A	–	–	–	(B, C)
B	A	B	–	–
C	A	–	C	–

	0	1	2
A	{ABC}	B	C
B	{ABC}	B	–
C	{ABC}	–	C

$\delta'(A, 0) = \epsilon$ closure $(\delta(\epsilon$ closure of $A, 0))$
$= \epsilon$ closure $((A, B, C), 0)$
$= \epsilon$ closure $(A, 0) \cup (B, 0) \cup (C, 0)$
$= \epsilon$ closure $- \cup A \cup A$
$= \epsilon$ closure $\{A\}$
$= \{A B C\}$

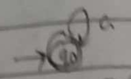
$\delta'(A, 1) = \epsilon$ closure $(\epsilon(\epsilon$ closure of $A, 1))$
$= \epsilon$ closure $((A B, C), 1)$
$= \epsilon$ closure $(A, 1) \cup (B, 1) \cup (C, 1)$
$= \epsilon$ closure $- \cup B \cup -$
$= \epsilon$ closure of $\{B\}$
$= \{B\}$

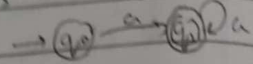
Q

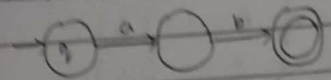


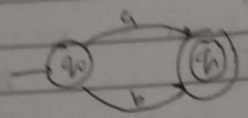
Date 5/3/25

Regular Expression
i) Kleen Closure (*)
ii) Positive closure (+)
iii) Concatenation (.) (a.b) आंद
iv) + Union (a+b) या

$a^* = \{\epsilon, a, aa, aaa, \ldots a^n\}$ 

$a^+ = \{a, aaaaa, \ldots\}$ 

a.b : 

a+b : 

a) 1.0.1 represented by $\{101\}$
b) abba $= \{abba\}$
c) As $\{10, 01\}$ is the union of $\{01\}$ & $\{10\}$
$= (01 + 10)$
d) The set $\{abb, a, b, bba\} = abb + a + b + bba$
e) $0^* = \{\epsilon, 0, 00, 000 \ldots\}$
f) $1.(1)^* = 1\{\epsilon, 1, 11, 111, 1111, \ldots\}$

Q Write a regular expression for the lang. accepting all the combination's of a's over the set where $\Sigma = \{a\}$
→ a^*

Q ... except ϵ → a^+

Q ~~a's b's~~ all possible combination of a's & b's
$R.E = (a+b)^* = \epsilon, a, b, aa, bb, ab, ba, \ldots$

$a^* + b^* = \epsilon, aaa, aa, aaa \ldots$
$\epsilon, b, bb, bbb \ldots$

$(a.b)^* = \epsilon, ab, abab, abab, \ldots$

Q Construct a R.E. for the language accepting all the string of 0 & 1 with ending with 00

00
100
000
1100
1000
0100
0000

$(0+1)^*(00)$
or
1. $R.E = (0+1)^* . 00$

Q R.E accepting all strings of 0's & 1's which start with 1 & end with 0.
10, 100, 110, 1000, 1
$R.E = 1.(1+0)^* . 0$

Q The lang starting & ending with a having any combination of b's
~~R.E = a.b*.a~~
a, aa, aba, abba, ...

$R.E = a.b^* a + a$

6/3/25

Q Write a R.E to denote a language & which accepts all the strings which begin or ends with either 00 or 11.

$((00) + (11)).(0+1)^* . ((00) + (11)) + (00+$

$R.E = ((00+11).(0+1)^*) + ((0+1)^*(00+11))$

Q. R.E denote a language L over Σ^* where $\Sigma = \{a, b\}$ such that 3^{rd} character from the right end of the string is always a.

$\Sigma^* = \Sigma : \{a, b\}^*$

_ a _ _

a a a

a b b

Smallest a a b

a b a

$R.E = (a+b)^* . a . (a+b)^* (a+b) (a+b)$

Q R.E for the lang L which accepts all the strings with atleast 2 b's over the $\Sigma = \{a, b\}$

bb

~~$(a+b)^* . bb . (a+b)^*$~~

$R.E = (a+b)^* b (a+b)^* b (a+b)^*$

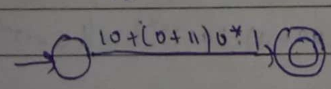
~~8/3/25~~

~~Q. Word~~

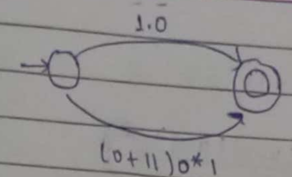
Direct Method of Regular Expression to finite automata

1) $\underbrace{10}_{R_1} + \underbrace{(0+11)0^*1}_{R_2}$

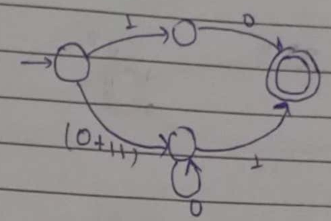
(a)



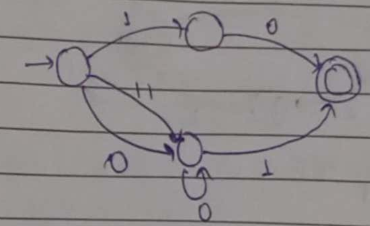
(b)



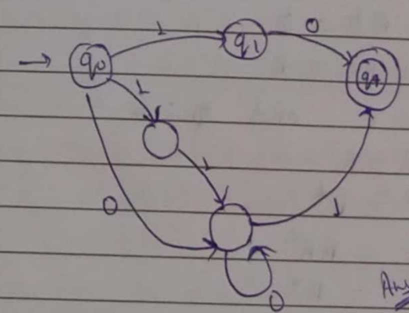
(c)



(d)



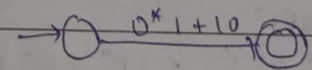
(e)



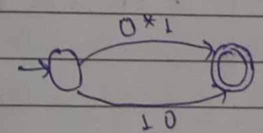
Ans

2.) $0^*1 + 10$

(a)



(b)

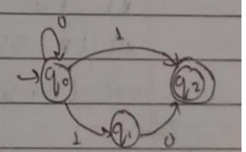


10

1

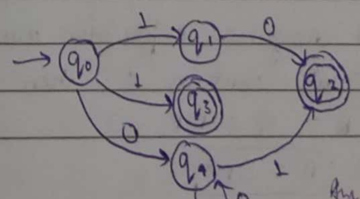
01

001



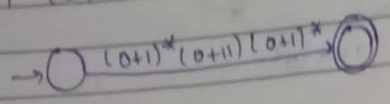
it is wrong bcz it can pass 010 which is incorr

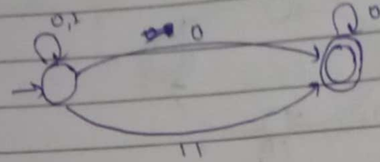
(c)

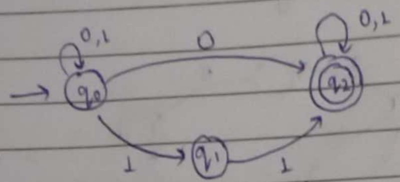


Ans

3.) $(0+1)^*(0+11)(0+1)^*$

a) 

b) 

c) 

Identities of RE

i) $\phi + R = R$

ii) $\phi R = R\phi = \phi$

iii) $\epsilon R = R\epsilon = R$

iv) $\epsilon^* = \epsilon$ and $\phi = \epsilon$

v) $R + R = R$

vi) $R^* R^* = R^*$

vii) $R.R^* = RR^*$

viii) $(R^*)^* = R^*$

ix) $\epsilon + RR^* = R^* = \epsilon R^* R$

x) $(PQ)^*P = P(QP)^*$

xi) $(P+Q)^* = (P^*Q^*)^* = (P^* + Q^*)^*$

xii) $(P+Q)R = PR + QR$

10/3/25

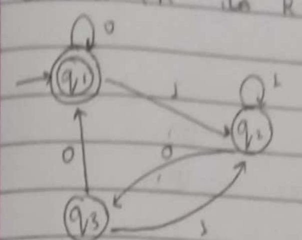
Arden's Theorem:

Let P & Q be regular expression over Σ, where $P \neq \epsilon$ does not contain ϵ, then

$R = Q + RP$

becomes $\boxed{R = QP^*}$

Q1 Convert FA to RE



$q_1 = q_1 0 + q_3 0 + \epsilon$ —(i)

$q_2 = q_1 1 + q_2 1 + q_3 1$ —(ii)

$q_3 = q_2 0$ —(iii)

Substitute value of q_3 in eq^n (ii)

$q_2 = q_1 1 + q_2 1 + q_2 01$

$\underset{R}{q_2} = \cancel{q_1 0}\ \underset{Q}{q_1 1} + \underset{R}{q_2}\underset{P}{(1+01)}$

$\cancel{q_3 = q_2 0}$

$\{R = Q + RP$, $R = QP^*\}$

$\boxed{q_2 = q_1 1 (1+01)^*}$ —(iv)

$q_3 = q_2 0 = q_1 1 (1+01)^* 0$ —(v)

Put q_3 in eq^n (i)

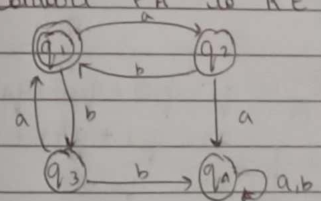
$q_1 = q_1 0 + q_1 1 (1+01)^* 00 + \epsilon$

$\underset{R}{q_1} = \underset{R}{q_1}\underset{P}{(0 + (1(1+01)^* 00))} + \underset{Q}{\epsilon}$ $\{R = RP + Q$, $R = QP^*\}$

$q_1 = \epsilon (0 + 1(1+01)^* 00)^*$ $[\epsilon . R = R]$

$q_1 = (0 + 1(1+01)^* 00)^*$

Q Convert FA to RE



$q_1 = q_2 b + q_3 a + \epsilon$ —(i)

$q_2 = q_1 a$ —(ii)

$q_3 = q_1 b$ —(iii)

$q_4 = q_2 a + q_3 b + q_4\, a+b$ —(iv)

Putting q_2 & q_3 in eq^n (i),

$q_1 = q_1 a.b + q_1 b.a + \epsilon$

$\underset{R}{q_1} = \underset{R}{q_1}\underset{P}{(ab + ba)} + \underset{Q}{\epsilon}$ $(R = RP + Q$, $R = QP^*)$

$q_1 = \epsilon . (ab + ba)^*$ $(\epsilon R = R)$

$q_1 = (ab + ba)^*$

Q

$q_1 = q_1 0 + \epsilon$ —— (i)

$q_2 = q_1 1 + q_2 1$ —— (ii)

$q_3 = q_2 0 + q_3 0{+}1$ —— (iii)

Take eq^n (i)

$\underset{R}{q_1} = \underset{R}{q_1}\,\underset{P}{0} + \underset{Q}{\epsilon}$ $\qquad R = RP + Q \Rightarrow R = QP^*$

$q_1 = \epsilon \cdot 0^*$ $\qquad \epsilon R = R$

$q_1 = 0^*$

For eq^n (ii), Put q_1 in eq^n (ii)

$\underset{R}{q_2} = \underset{Q}{0^* 1} + \underset{R}{q_2}\,\underset{P}{1}$ $\qquad R = RP + Q = QP^*$

$q_2 = 0^* 1 . 1^*$ $\qquad RR^* = R^*$

$q_2 = 0^* 1^*$

because of two final state

$RE = q_1 + q_2$

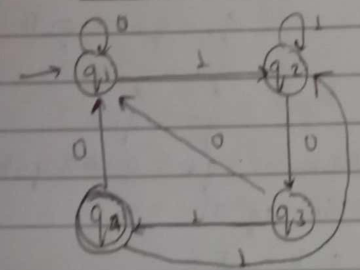
$= 0^* + 0^* 1^*$

$= 0^* (\epsilon + 1^*)$ $\qquad (\epsilon + R^* = R^*)$

$\boxed{RE = 0^* 1^*}$

12/3/25

Q



$q_1 = q_1 0 + q_3 0 + q_4 0 + \epsilon$

$q_2 = q_1 1 + q_2 1 + q_4 1$

$q_3 = q_2 0$

$q_4 = q_3 1$

Put $q_3 = q_2 0$ in eq^n (iv)

$q_4 = q_2 0.1$

Put $q_4 = q_2 0 1$ in eq^n (ii)

$q_2 = q_1 1 + q_2 1 + q_4 1$

$q_2 = q_1 1 + q_2 1 + q_2 0 1 . 1$

$\underset{R}{q_2} = \underset{Q}{q_1 1} + \underset{R}{q_2}\,\underset{P}{(1 + 011)}$ $\qquad R = RP + Q = QP^*$

$q_2 = q_1 1 (1 + 011)^*$ —— (vi)

Put $q_3 = q_2 0$ & $q_4 = q_2 0 1$ in eq^n (iv)

$q_1 = q_1 0 + q_3 0 + q_4 0 + \epsilon$

$\Rightarrow q_1 = q_1 0 + q_2 0.0 + q_2 0.1.0 + \epsilon$

$\Rightarrow q_1 = q_1 0 + q_2 (00 + 010) + \epsilon$

From eq^n (vi).

$q_1 = q_1 0 + q_1 1 (1 + 011)^* (00 + 010) + \epsilon$

$\cancel{q_1 = q_1 0 + q_3 0 + q_4 0 + \epsilon}$

$\cancel{= q_1 0 + q_2 0.0 + q_2 0 1 . 0 + \epsilon}$

$\underset{R}{q_1} = \underset{R}{q_1}\,\underset{P}{(0 + 1 (1 + 011)^* (00 + 010))} + \underset{Q}{\epsilon}$ $\qquad [R = RP + Q = QP^*]$

$q_1 = \epsilon (0 + 1 (1 + 011)^* (00 + 010))^*$ $\qquad (\epsilon . R = R)$

$q_1 = (0 + 1 (1 + 011)^* (00 + 010))^*$

Put q_1 in eq^n (vi)

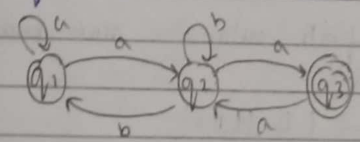
$q_2 = (0 + 1 (1 + 011)^* (00 + 010))^*) 1 (1 + 011)^*$

Put q_2 in eq^n (v)

$q_4 = (0 + 1 (1 + 011)^* (00 + 010))^*) 1 (1 + 011)^* 0 1$

17/3/25

Q



$q_1 = q_1 a + q_2 b + \epsilon$ —— (i)

$q_2 = q_1 a + q_2 b + q_3 a$ —— (ii)

$q_3 = q_2 a$ —— (iii)

Putting $q_3 = q_2 a$ in eq^n (ii)

$q_2 = q_1 a + q_2 b + q_2 a \cdot a$ $\quad R = RP + Q$

$\underset{R}{q_2} = \underset{Q}{q_1 a} + \underset{R}{q_2}\underset{P}{(b + aa)}$ $\quad R = QP^*$

$q_2 = q_1 a(b + aa)^*$ ——— (iv)

Putting $q_2 = q_1 a(b+aa)^*$ in eq^n (i)

$q_1 = q_1 a + q_1 a(b+aa)^* b + \epsilon$

$\underset{R}{q_1} = \underset{R}{q_1}\underset{P}{(a + a(b+aa)^* b)} + \underset{Q}{\epsilon}$ $\quad R = RP + Q$, $\Rightarrow R = QP^*$

$q_1 = \epsilon \cdot (a + a(b+aa)^* b)^*$ $\quad (\epsilon R = R)$

$q_1 = (a + a(b+aa)^* b)^*$ ——— (v)

Put eq^n (v) in eq^n (iv)

$q_2 = (a + a(b+aa)^* b)^* a(b+aa)^*$

Put q_2 in q_3 (eq^n (iii))

$q_3 = (a + a(b+aa)^* \cdot b)^* \cdot a(b+aa)^* a$

2 way finite automata

The 2 way finite automata is a model in which linear direction is mentioned on receiving particular input & being in same current state, there are 2 directions allowed these are left & right direction.

The 2 way finite automata also have 5 tuple $(Q, \Sigma, q_0, F, \delta)$, i/p, o/p

$\delta : Q \times \Sigma \rightarrow (Q, \{L, R\})$

δ \ I/P	0	1
$\rightarrow q_0$	(q_0, R)	(q_1, R)
q_1	(q_2, R)	(q_2, L)
q_2	(q_3, R)	(q_2, R)
q_3	(q_3, R)	(q_3, R)

String : 001101

$q_0 R \rightarrow q_0 R \rightarrow q_1 R \rightarrow q_2 L \rightarrow q_3 R \rightarrow q_3 R$

$\delta(q_0\ 001101) \vdash 0 q_0 01101$
$\vdash 00 q_0 1101$
$\vdash 001 q_1 101$
$\vdash 00 q_2 1101$
$\vdash 001 q_2 101$
$\vdash 0011 q_2 01$
$\vdash 00110 q_2 1$
$\vdash 001101 q_3$

20/9/25
ASS-1

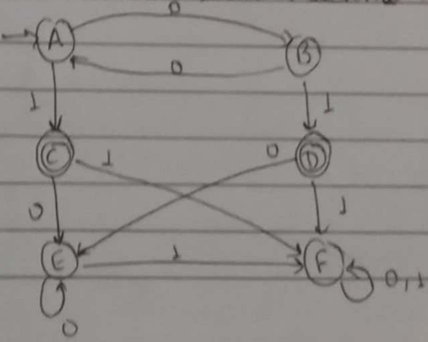
Q1. Discuss finite automata with the help of appropriate example.

Q2. Make a finite state m/c for any decimal no. which is divisible by 3.

Q3. Construct moore m/c equivalent to the given mealy m/c.

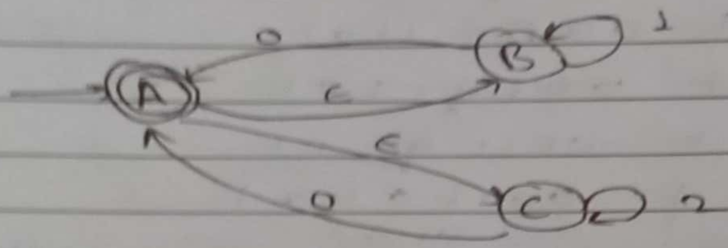
	0		1	
State	State	O/P	State	O/P
q_1	q_1	1	q_2	0
q_2	q_4	1	q_4	1
q_3	q_2	1	q_3	1
q_4	q_3	0	q_1	1

Q4. How to reduce the no. of state of DFSM (Deterministic Finite State Machine), Minimization



Construct a DFA for a language which starts from a and ends with a where $\Sigma = \{a, b\}$

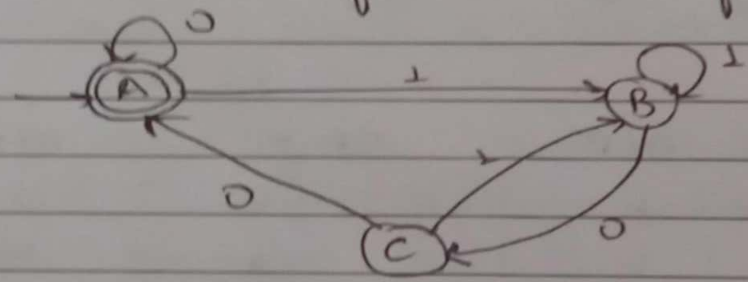
Convert the following into NFA without ε moves.



Convert the following regular expression to NFA & then convert into DFA:

$$RE = (a+b)^* ab$$

Write RE for the following DFA:



bmission → 27 Mar, 2025